

Amr Mahmoud

Software Engineer | Backend Engineer

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Summary

Back-End Engineer and Computer Science student with experience building RESTful APIs, authentication systems, and database-driven backends using Node.js, Laravel, and Flask. Strong foundation in data structures, system design, and Linux-based development environments.

Experience

CS-Club Vice-President & Instructor: *E-JUST Clubs* February 2026 – present

- Taught Version Control (Git&GitHub) for three lectures that helped students understand basic Git command and how to collaborate with others via GitHub

Android Development Trainee: *Digital Egypt Pioneers Initiative (DEPI)* July 2025 – November 2025

- Developed native Android applications using **Kotlin** and **Jetpack Compose**.
- Worked within MVVM-based Android architecture, integrating backend APIs.

IT Field Intern: *Egypt Japan University of Science and Technology (E-JUST)* July 2025 – September 2025

- Developed backend solutions using PHP Laravel framework for university website
- Implemented backend features for the university course catalogue using Laravel
- Developed Student Affairs portal website using Laravel & React and Integration with Paymob Payment gateway.
- Gained practical understanding of networking fundamentals (TCP/IP, routing, basic security concepts) relevant to backend.

Education

Bachelor of Science in Computer Science and Information Technology

Egypt Japan University of Science and Technology (E-JUST)

2023 – 2027 (Expected)

- GPA: 3.51/4.0
- Current Level: 3rd Year
- Full Scholarship Recipient through Innovators Support Fund (**ISF**)
- CS-Club Vice president

Projects

E-JUST SIS:

- Designed and implemented the application phase of the SIS (the project still under active development).
- Architected a Node.js/PostgreSQL admissions backend managing 90+ REST API endpoints across 15 modules
- Secured multi-role access (Staff, Staff-Admin, Student, IT-Admin) using JWT-based RBAC.
- **Technologies:** Node.js, Express.js, PostgreSQL, JWT, API Dog, BullMQ, zod

Green Rabbit:

- Integrated third-party AI services via the Perplexity AI API to power an intelligent chatbot.
- Secured API endpoints with JWT authentication and applied rate limiting through Redis
- Documented the RESTful interfaces using API Dog to streamline frontend integration.
- **Technologies:** Node.js, Express.js, PostgreSQL, JWT, API Dog, BullMQ, Perplexity AI, cron jobs, railway

Campus Connect: [GitHub](#)

- Developed a cross-platform university engagement portal for E-JUST students featuring event management, club discovery, room booking, facility reservations, and social feed with posts and interactions.
- Designed relational database schema and implemented secure authentication and authorization flows used by web and mobile clients.
- Built RESTful API with Express.js implementing JWT authentication, (RBAC) Role-Based Access Control, and layered architecture (Controllers, Services, Repositories) connected to MariaDB.
- **Technologies:** Node.js, Express.js, MariaDB, zod JWT, API Dog, PM2

Skills

Programming Languages: Python, C++, PHP, MATLAB, Java, Kotlin, JavaScript

Web Technologies: Flask, Laravel, Node.js, Express.js, HTML, CSS

Tools & Platforms: Git, GitHub, Linux, VS Code, IntelliJ IDEs, Docker, Neo Vim, SQL, NoSQL, API-Dog, JEST

Core Concepts: Object-Oriented Programming, Data Structures, Algorithms, Problem Solving, API, Software Development Lifecycle, System Design

Languages: Arabic (Native), English (Upper Intermediate)